

Nirupam Gupta

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Education

Doctor of Philosophy (Ph.D.)	2019
<i>Mechanical Engineering, University of Maryland, College Park, USA</i>	
Bachelors in Technology (B.Tech)	2013
<i>Electrical Engineering, Indian Institute of Technology, Delhi, India</i>	

Employment

Tenure-track Assistant Professor	2024 -
<i>Department of Computer Science, University of Copenhagen, Denmark</i>	
Postdoctoral Researcher	2021 - 2024
<i>School of Computer Science, EPFL, Switzerland</i>	
Postdoctoral Researcher	2019 - 2021
<i>Department of Computer Science, Georgetown University, USA</i>	

Fundings

Interaction Between Privacy and Robustness in Distributed Learning [Co-PI].	2025 - 26
<i>Received 14,000 € by CNRS, France for international emerging action.</i>	
TruBrain: Trustworthy Distributed Brain-inspired Systems [Co-PI].	2023 - 24
<i>Received 522,452 CHF (550,000 €) by the Swiss NSF (National Science Foundation) through CHIST-ERA ERA-NET 2022 call. Main PI: Ihcen Alouani, Queen's University, Belfast.</i>	

Research & Teaching Activities

My main area of research is distributed machine learning (a.k.a. federated learning), with focus on trustworthiness - robustness and privacy. I have published over 40 peer-reviewed papers on these and related topics, including a textbook on Robust Machine Learning. Complete list of my publications can be found on my [Google scholar page](#).

I currently teach the following courses at the University of Copenhagen:

1. **Machine Learning B (MLB)**. The course provides an introduction to optimization principles and generalization theory that are essential to ML.
2. **Privacy in Machine Learning (PriMaL)**. The course introduces the concept of differential privacy and decentralization learning, both of which are state-of-the-art techniques for preserving privacy of training data in ML.
3. **Robust Machine Learning (RoMaL)**. The course introduces the fundamentals of robustness against strong (data or model) contamination in machine learning.

Research Awards

Best Paper, International Conference on Distributed Computing and Networking (ICDCN)	2023
Best Paper Runner-up, International Symposium on Reliable Distributed Systems (SRDS)	2022

Ph.D. Co-supervision

Mingzhi Wang (co-supervisor: Prof. Yevgeny Seldin), <i>Dept. of Computer Science, University of Copenhagen, Denmark</i>	2025 -
Thomas Boudou (co-supervisors: Dr. Aurélien Bellet & Dr. Batiste Le Bars), <i>INRIA-Inserm, University of Montpellier, France</i>	2024 -
Dr. John Stephan. (co-supervisors: Prof. Rachid Guerraoui & Dr. Rafael Pinot), <i>School of Computer Science, EPFL, Switzerland</i>	2021 - 25
Dr. Sadegh Farhadkhani. (co-supervisors: Prof. Rachid Guerraoui & Dr. Rafael Pinot), <i>School of Computer Science, EPFL, Switzerland</i>	2021 - 25
Dr. Youssef Allouah. (co-supervisors: Prof. Rachid Guerraoui & Dr. Rafael Pinot), <i>School of Computer Science, EPFL, Switzerland</i>	2022 - 24
Dr. Shuo Liu. (co-supervisor: Prof. Nitin H. Vaidya), <i>Dept. of Computer Science, Georgetown University, USA.</i>	2021 - 24

M.Sc. Supervision

I have supervised 3 Msc theses in the Department of Computer Science at University of Copenhagen (DIKU), since 2024. I have additionally co-supervised 4 other MSc theses: 2 at DIKU, and 2 at the Technical University of Denmark (DTU).

Community services, organizations and outreach

Affiliations. Pioneer Center for AI, the Denmark Learning Theory and Applications (DeLTA) group, and the European Lab for Learning & Intelligent Systems (ELLIS).

Program co-chair -

International Conference on Networked Systems (NETYS), Rabat, Morocco	May, 2024
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Senior program committees (Area chair) -

Annual Conference on Artificial Intelligence and Statistics (AISTATS)	2027
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Program committees -

International Conference on Machine Learning (ICML)	2023 - 27
International Conference on Learning Representations (ICLR)	2023 - 27
Annual Conference on Neural Information Processing Systems (NeurIPS)	2023 - 27
Annual Conference on Artificial Intelligence and Statistics (AISTATS)	2023 - 27
AAAI International Conference on Artificial Intelligence	2026 - 27
ACM Conference on Computer and Communications Security (CCS)	2026 - 27
IEEE Secure and Trustworthy Machine Learning (SaTML)	2025 - 27
IEEE International Conference on Distributed Computing Systems (ICDCS)	2024, 26
Symposium on Reliable Distributed Systems (SRDS)	2023

Reviewer for Journal of Machine Learning Research (JMLR), Transactions on Machine Learning Research (TMLR), International Conference on Machine Learning (ICML).

Co-organized workshops -

3rd Workshop on Machine Learning Theory at D3A Conference, Denmark	Oct., 2026
Workshop on Learning Theory, Univeristy of Copenhagen, Denmark	June, 2026
Workshop on Fundamentals of Security and Privacy in AI at Swiss AI Days Conference, Switzerland	March, 2026
2nd Workshop on Machine Learning Theory at D3A Conference, Denmark	Aug., 2025
3rd Workshop on the Principles of Distributed Learning at PODC, France	June, 2024
2nd Workshop on the Principles of Distributed Learning at DISC, Italy	Oct., 2023
1st Workshop on the Principles of Distributed Learning at PODC, Italy	July, 2022

Invited Talks (last 4 years) -

Robust Federated Learning. <i>At NEO-Finality PhD Summer School, INRIA Sophia-Antipolis, Nice, France</i>	Sept., 2026
Robust Distributed Learning. <i>At Learning Theory Summer School, Copenhagen</i>	June, 2026
Robust Distributed Learning. <i>At SPTDC PhD Summer School, Heraklion, Greece</i>	April, 2026
Robust Federated Learning. <i>at Core-AI spring school, Montpellier, France</i>	March, 2026
Embedding Safety in AI by Robustifying Model Training. <i>At A2S2 PhD Seasonal School, Copenhagen, Denmark</i>	Jan., 2026
Machine Learning in Untrusted Environments. <i>At Northeastern University, Rutgers University and University of Maryland - College Park, USA.</i>	Jul-Aug, 2025
Machine Learning in Untrusted Distributed Environment. <i>At the 33rd European Conference on Operational Research (EURO), Copenhagen, Denmark.</i>	July, 2024
Machine Learning in Untrusted Environment. <i>At INRIA Montpellier (France), INRIA Sophia-Antipolis (Nice, France) and University of Copenhagen (Denmark).</i>	Dec., 2023
Byzantine Machine Learning. <i>At the International Symposium on Distributed Computing (DISC'23), Italy.</i>	Oct., 2023
Distributed Learning with Adversarial Nodes. <i>At the GDR RSD Summer School on Distributed Learning, INRIA & CNRS Lyon, France.</i>	Sept., 2023

Publications

Author ordering: Authors are listed alphabetically (as per last name). Asterisk * and dagger † on my name indicates leading authorship and supervisory role, respectively.

Books and chapters

Book: Robust Machine-Learning, Distributed Methods for Safe AI
Rachid Guerraoui, **Nirupam Gupta***, Rafael Pinot. *Springer Nature*, 2024

Chapter: Robustness & Privacy in Federated Learning
Rachid Guerraoui and **Nirupam Gupta***.

Large Language Models and Cybersecurity: Trends in risk, exposure and mitigation. *Springer*, 2024

Journal publications

1. On The Relevance of Byzantine Robust Optimization Against Data Poisoning
Sadegh Farhadkhani, Rachid Guerraoui, **Nirupam Gupta***, Rafael Pinot. *Journal of Machine Learning (JMLR)*, 2026.

2. Byzantine Machine Learning: A Primer
Rachid Guerraoui, **Nirupam Gupta**^{*}, Rafael Pinot. *ACM Computing Surveys*, 2023.
3. Byzantine Fault-Tolerance in Federated Local SGD under 2f-Redundancy
Nirupam Gupta^{*}, Thinh T. Doan, and Nitin H. Vaidya. *IEEE Transactions on Control of Network Systems*, 2023.
4. [On Pre-Conditioning of Decentralized Gradient-Descent when Solving a System of Linear Equations](#)
Kushal Chakrabarti, **Nirupam Gupta**[†], and Nikhil Chopra. *IEEE Transactions on Control of Network Systems*, 2022.
5. [Iterative Pre-Conditioning for Expediting the Distributed Gradient-Descent Method: The Case of Linear Least-Squares Problem](#)
Kushal Chakrabarti, **Nirupam Gupta**^{*}, and Nikhil Chopra. *Automatica*, 2022.
6. [Robustness of Iteratively Pre-Conditioned Gradient-Descent Method: The Case of Distributed Linear Regression Problem](#)
Kushal Chakrabarti, **Nirupam Gupta**, and Nikhil Chopra. *IEEE Control Systems Letters*, 2021.
7. [Preserving Statistical Privacy in Distributed Optimization](#)
Nirupam Gupta^{*}, Shripad Gade, Nikhil Chopra, and Nitin H. Vaidya. *IEEE Control Systems Letters*, 2021.
8. False Data Injection Attacks in Bilateral Teleoperation Systems
Yimeng Dong, **Nirupam Gupta**, and Nikhil Chopra. *IEEE Transactions on Control Systems Technology*, 2018.
9. Content Modification Attacks on Consensus Seeking Multi-Agent System with Double-Integrator Dynamics
Yimeng Dong, **Nirupam Gupta**, and Nikhil Chopra. *AIP Chaos - Journal of Nonlinear Science*, 2016.

Conference proceedings

1. Tight Stability Bounds for Robust Distributed Learning: Byzantine Failures Hurt Generalization More than Data Poisoning
Thomas Boudou, Batiste Le Bars, **Nirupam Gupta**[†], Aurélien Bellet. *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026.
2. Robust Federated Inference
Akash Dhasade, Sadegh Farhadkhani, Rachid Guerraoui, **Nirupam Gupta**^{*}, Maxime Jacovella, Anne-Marie Kermarrec, Rafael Pinot. *Proceedings of the 29th International Conference on Learning Representations (ICLR)*, 2026.
3. Reconciling Communication Compression and Byzantine-Robustness in Distributed Learning
Diksha Gupta, Antonio Honsell, Chuan Xu, **Nirupam Gupta**^{*} and Giovanni Neglia. *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026.
4. Adaptive Gradient Clipping for Robust Federated Learning
Youssef Allouah, Rachid Guerraoui, **Nirupam Gupta**^{*}, Ahmed Jellouli, Geovani Rizk, and John Stephan. *International Conference on Learning Representations (ICLR)*, 2025 [Spotlight, acceptance rate of 5%].
5. Revisiting Ensembling in One-Shot Federated Learning
Youssef Allouah, Akash Dhasade, Rachid Guerraoui, **Nirupam Gupta**, Anne-Marie Kermarrec, Rafael Pinot, Rafael Pires, Rishi Sharma. *In the 38th Conference on Neural Information Processing Systems (NeurIPS)*, 2024.

6. Fine-Tuning Personalization in Federated Learning to Mitigate Adversarial Clients
Youssef Allouah, Abdellah El Mrini, Rachid Guerraoui, **Nirupam Gupta**[†] and Rafael Pinot. *In the 38th Conference on Neural Information Processing Systems (NeurIPS), 2024.*
7. Tackling Byzantine Clients in Federated Learning
Youssef Allouah, Sadegh Farhadkhani, Rachid Guerraoui, **Nirupam Gupta**, Rafael Pinot, Geovani Rizk, and Sasha Voitovich. *Proceedings of the 41st International Conference on Machine Learning (ICML), 2024.*
8. Robust Distributed Learning: Tight Error Bounds and Breakdown Point under Data Heterogeneity
Youssef Allouah, Rachid Guerraoui, **Nirupam Gupta**[†], Rafael Pinot, and Geovani Rizk. *In the 37th Conference on Neural Information Processing Systems (NeurIPS), 2023* [Spotlight, acceptance rate of 5%].
9. On the Privacy-Robustness-Utility Trilemma in Distributed Learning
Youssef Allouah, Rachid Guerraoui, **Nirupam Gupta**[†], Rafael Pinot, and John Stephan. *Proceedings of the 40th International Conference on Machine Learning (ICML), 2023.*
10. Robust Collaborative Learning with Linear Gradient Overhead
Sadegh Farhadkhani, Rachid Guerraoui, **Nirupam Gupta**^{*}, Lê-Nguyen Hoang, Rafael Pinot, and John Stephan. *Proceedings of the 40th International Conference on Machine Learning (ICML), 2023.*
11. Fixing by Mixing: A Recipe for Optimal Byzantine ML under Heterogeneity
Youssef Allouah, Sadegh Farhadkhani, Rachid Guerraoui, **Nirupam Gupta**[†], Rafael Pinot, and John Stephan. *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS), 2023.*
12. Impact of Redundancy on Resilience in Distributed Optimization and Learning
Shuo Liu, **Nirupam Gupta**[†], and Nitin H. Vaidya. *Proceedings of the 24th International Conference on Distributed Computing and Networking (ICDCN) [Best Paper], 2023.*
13. Democratizing Machine Learning: Resilient Distributed Learning with Heterogeneous Participants
Karim Boubouh, Amine Boussetta, **Nirupam Gupta**, Alexandre Maurer, and Rafael Pinot. *Proceedings of the 41st International Symposium on Reliable Distributed Systems (SRDS), 2022.*
14. Byzantine Machine Learning Made Easy by Resilient Averaging of Momentums
Sadegh Farhadkhani, Rachid Guerraoui, **Nirupam Gupta**^{*}, Rafael Pinot, and John Stephan. *Proceedings of the 39th International Conference on Machine Learning (ICML), 2022.*
15. Accelerating Distributed SGD for Linear Regression using Iterative Pre-Conditioning
Kushal Chakrabarti, **Nirupam Gupta**, and Nikhil Chopra. *Proceedings of the 3rd Conference on Learning for Dynamics and Control (L4DC), 2021.*
16. Byzantine Fault-Tolerance in Decentralized Optimization under 2f-Redundancy
Nirupam Gupta^{*}, Thanh T. Doan, and Nitin H. Vaidya. *The 2021 American Control Conference (ACC).*
17. Differential Privacy and Byzantine Resilience in SGD: Do They Add Up?
Rachid Guerraoui, **Nirupam Gupta**^{*}, Rafaël Pinot, Sébastien Rouault, and John Stephan. *The ACM Symposium on Principles of Distributed Computing (PODC), 2021.*
18. Approximate Byzantine Fault-Tolerance in Distributed Optimization
Shuo Liu, **Nirupam Gupta**[†], and Nitin H. Vaidya. *The ACM Symposium on Principles of Distributed Computing (PODC), 2021.*

19. Preserving Statistical Privacy in Distributed Optimization
Nirupam Gupta*, Shripad Gade, Nikhil Chopra, and Nitin H. Vaidya. *The 59th IEEE Conference on Decision and Control (CDC)*, 2020.
20. Fault-Tolerance in Distributed Optimization: The Case of Redundancy
Nirupam Gupta*, and Nitin H. Vaidya. *The ACM Symposium on Principles of Distributed Computing (PODC)*, 2020.
21. Iterative Pre-Conditioning to Expedite the Gradient-Descent Method
 Kushal Chakraborty, **Nirupam Gupta**, and Nikhil Chopra. *The 2020 American Control Conference (ACC)*.
22. On Distributed Solution of Ill-Conditioned System of Linear Equations under Communication Delays
 Kushal Chakraborty, **Nirupam Gupta**, and Nikhil Chopra. *The Dec'19 Indian Control Conference (ICC)*.
23. Byzantine Fault-Tolerant Parallelized Stochastic Gradient Descent for Linear Regression
Nirupam Gupta, and Nitin Vaidya. *The 2019 Allerton Conference at UIUC*.
24. Statistical Privacy in Distributed Average Consensus: Bounded Real Inputs
Nirupam Gupta*, Jonathan Katz, and Nikhil Chopra. *The 2019 American Control Conference (ACC)*.
25. Model-Based Encryption: Privacy of States in Networked Control Systems
Nirupam Gupta*, and Nikhil Chopra. *The 2018 Allerton Conference at UIUC*.
26. Privacy in Distributed Average Consensus
Nirupam Gupta*, Jonathan Katz, and Nikhil Chopra. *The World Congress of IFAC*, 2017.
27. Robustness of distributive double-integrator consensus to loss of graph connectivity
Nirupam Gupta*, Yimeng Dong, and Nikhil Chopra. *The 2017 American Control Conference (ACC)*.
28. Confidentiality in Distributed Average Information Consensus
Nirupam Gupta*, and Nikhil Chopra. *The 55th IEEE Conference on Decision and Control (CDC) 2016*.
29. On Content Modification Attacks in Bilateral Teleoperation Systems
 Yimeng Dong, **Nirupam Gupta**, and Nikhil Chopra. *The 2016 American Control Conference (ACC)*.
30. Stability analysis of a two-channel feedback networked control system
Nirupam Gupta*, and Nikhil Chopra. *The 2016 Indian Control Conference (ICC)*.